Computer Communications 154 (2020) 455–464

Contents lists available at ScienceDirect

Computer Communications

journal homepage: www.elsevier.com/locate/comcom

A lightweight authentication and key agreement scheme for Internet of Drones

Yunru Zhang [a,b], Debiao He [a,b,*], Li Li [a], Biwen Chen [c]

[a] Key Laboratory of Aerospace Information Security and Trusted Computing, Ministry of Education, School of Cyber Science and Engineering, Wuhan University, Wuhan 430072, China

[b] Cyberspace Security Research Center, Peng Cheng Laboratory, Shenzhen 518055, China

[c] School of Computer Science, Wuhan University, Wuhan 430072, China

ARTICLE INFO

Keywords:
Internet of Drones
Lightweight
Authentication and key agreement

ABSTRACT

Drones in Internet of Drones (IoD) can be able to reconnoiter environment, transport the commodity with the help of embedded various sensors. They have been widely used in various fields and brought a great convenience to the production and life. But data collected by sensors embedded in drones are facing new security challenges and privacy issues with the technology update over time. For the sake of ensuring the security of transmitted data, many authentication and key agreement (AKA) schemes have been proposed in the past. Nevertheless, most of schemes are subjected to serious security risks and have high communication and computation cost. To address these issues in IoD, we propose a lightweight AKA scheme in which there are only secure one-way hash function and bitewise XOR operations when drones and users mutually authenticate each other. The proposed scheme can achieve AKA-security under the random oracle model and withstand various known attacks. Meanwhile, the security comparison demonstrates our proposed scheme provides better security. In terms of communication and computation cost, our proposed scheme has better functionality features than the other two schemes.

1. Introduction

Internet of Drones (IoD) [1] has been widely used in various fields and brought a great convenience to the production and life of people owing to their kinds of sensor equipment [2,3]. For instance, it has been used in military reconnaissance, logistics transportation and disaster relief. Fig. 1 exhibits a typical drone application for surveillance. The sensors embedded in drone can collect and analyze the physical phenomena (e.g. humidity, temperature, atmospheric pressure), and also the embedded camera and microphone can transmit video back to controller via wireless communication technology (e.g. WiFi, bluetooth). Thus, controller can get real-time information through controlling drones at a distance.

The mobility of drones makes them more widely used in Internet of Things (IoT) environment [4,5]. But drone's weight, energy source and communication technology are the main factors impacting its service, such as communication method, flight range, flight endurance, load capacity and so on [6–8]. Therefore, different tasks may need different sensors combination types. For example, drones used for disaster relief should be equipped with infrared detectors and cameras, but drones used for aerial photography work mainly with cameras and microphones, and drones used for logistics transportation shall have positioning system. In addition, multi-drones can collect data in a distributed manner, meanwhile, the clustered working model can prolong sensor devices' battery life and reduce the cost of deploying the infrastructure [9,10].

Data collected by sensors embedded in drones are facing new security challenges and privacy issues with the technology update over time. The collected data may contain highly sensitive information (e.g. for military) and the fragile communication network among drones makes the transmitted data intercepted and captured easily. Mutual authentication can be used to verify the real identities of the communication participants before sharing secrets without sending sensitive information via insecure channel [11,12]. The AKA scheme can achieve this goal, which generates a shared session key to encrypt the subsequent communication messages. Therefore, the drones and users can mutual authenticate each other, users with session key can get the collected data but others cannot.

There are several factors must be paid attention to when design an AKA scheme, owing to the resource constrained drone device. On the one side, a more powerful adversary would make the protection mechanism based on the assumption that any adversary is incapable

* Corresponding author at: Key Laboratory of Aerospace Information Security and Trusted Computing, Ministry of Education, School of Cyber Science and Engineering, Wuhan University, Wuhan 430072, China.
E-mail address: hedebiao@163.com (D. He).

https://doi.org/10.1016/j.comcom.2020.02.067
Received 7 December 2019; Received in revised form 25 January 2020; Accepted 23 February 2020
Available online 29 February 2020
0140-3664/© 2020 Elsevier B.V. All rights reserved.



Fig. 1. The typical application of IoD.

of calculating the solution to a specific mathematical problem insecure. On the other side, with the limited resources, drones cannot execute complex operations on large datasets [13,14]. Like, the operations drones execute in the authentication phase should be sufficiently lightweight. In such that case, it is critical to achieve authentication between drones and users (controllers) before sharing the collected data, which also satisfies confidential requirement simultaneously.

In existing literatures, symmetric cryptographic was usually used to implement lightweight authentication scheme. However, it does not support user anonymity. Subsequently, public key infrastructure (PKI) was also pointed out unsuitable for the IoD environment owing to its complex certificate management. Identity-based cryptographic (IBC), with user's identity (e.g. email address, phone number) being his/her public key, is probably the appropriate option.

The key contributions in this paper are listed as follows:

- We propose a lightweight and efficient AKA scheme for the IoD architecture, in which there are only secure one-way hash function and bitewise XOR operations.
- The proposed scheme can satisfy mutual authentication and AKA-secure by means of provable security, and can withstand various known attacks through informal security analysis. The security comparison demonstrates our proposed scheme provides better security.
- In terms of communication and computation cost, the proposed scheme have better functionality features than the other two schemes in [15,16].

The remaining parts of this paper is organized as follows. We review some related literature on existing AKA schemes in Section 2. Section 3 describes the network model and the security requirements that proposed scheme needs to meet. In Section 4 we depict the proposed AKA scheme, whose security analysis is described in Section 5. We compare our proposed scheme with schemes in [15,16] in terms of communication cost and computation cost in Section 6. Section 7 makes a conclusion of the paper.

2. Related work

AKA scheme allows participants to generate a common session key via an insecure channel when they mutual authenticate each other. The scheme of remote authentication on the basis of password was introduced by Lamport for the first time [17], in the whole scheme only one-way hash function was needed in the whole scheme. Inspiring from this seminal work, many more secure authentication schemes and analysis schemes were come up with the innovative proposals in various environments [18–21].

Turkanovic et al. [22] was the first to put forward a novel AKA scheme between users and nodes without the help of gateway node. The scheme is befitting the resource-limited nodes as the use of hash function and bitewise XOR operation. However, Farash et al. [23] pointed out Turkanovic et al. s' scheme cannot resist against man-in-the-middle attack, node impersonation attack, and also cannot provide nodes anonymity and user traceability. Farash et al. proposed a new and improved AKA scheme to overcome the drawbacks in Turkanovic et al. s' scheme.

Unfortunately, Amin et al. [24] also found some security weaknesses of Farash et al. s' scheme, such as known specific temporary information attack, off-line password guessing attack, user impersonation attack and so on. Amin et al. designed a robust AKA scheme based on smart card. Later, Amin et al. s' scheme suffered from smart card lost attack and off-line password guessing attack were come up by Jiang et al. [25]. Challa et al. [26] put forward a new signature based AKA scheme using the elliptic curve cryptography. Along with the security of the scheme is the increased communication and computation overhead compared with other not using elliptic curve cryptography.

However, there are certificate management problem and key escrow problem in traditional PKI and IBC respectively. In order to address these issues, also consider that the execution time of pair operation [27,28] is much larger than other standard operations, several certificate-less public key cryptography (CL-PKC) AKA schemes based on pairing-free were introduced [29–32]. Nevertheless, neither of them were proved to be secure. Thereafter, Seo et al. [33] first put forward a pairing-free certificate-less signcryption tag key encapsulation mechanism (CLSC-TKEM). However, neither the existing CL-AKA schemes or CLSC-TKEM schemes have resolved user revocation issues. That means, once an adversary capture a drone, it can access all the information no matter whether it is already collected or about to be collected.

For the sake of revoking a compromised drone to protect the whole network, Won et al. [34] pointed out an efficient and secure certificate-less scheme for the drones. They proposed corresponding schemes considering the three different communication scenarios of drones. The first scenario is one-to-one, the authors proposed a CLSC-TKEM which could provide mutual authentication and key agreement and satisfy user revocation. For the next one-to-many scenario, they put forward a multi-recipient encryption scheme through which drones could share sensitive data with multiple smart devices. And for the last many-to-one, a certificate-less data aggregation scheme would allow drones collect data from numerous smart devices.

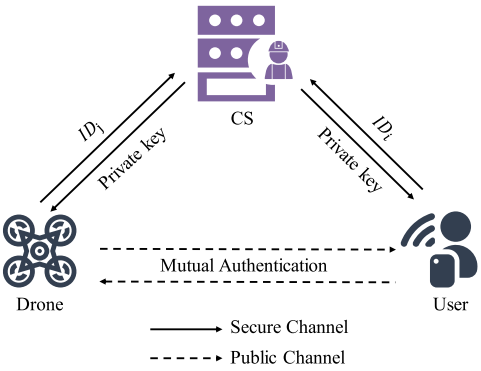


Fig. 2. The network model of designed framework.

3. System model

3.1. Network model

The network model of designed framework is described in Fig. 2 which contains three participants: Control Server(CS), mobile users(U_i) and drones(V_j).

1. CS: It is considered as a trusted party and responsible for registering every user and drone. CS generates long term secret keys of U_i and V_j according to their identities.
2. U_i: The user having a smart device (e.g. smart phone) gets his/her secret key form CS in registration phase. Before accessing and communicating with drones on the mission, he/she should be verified.
3. V_j: The drones also get their secret keys from CS in the registration phase. After verifying U_i's validity, V_j and U_i establish a session key to make sure the security of communication.

3.2. Security requirements

In the light of the intrinsic characteristics of authentication scheme for IoD architecture, our proposed AKA scheme should meet the following security requirements [35–40].

- *Mutual Authentication.* To ensure the validity of participants and their received messages, users and drones should be capable of authenticating the integrity and timeliness of identities transmitted transcripts.
- *Anonymity.* The scheme should guarantee the entities' identities privacy. No one else can get their real identities except the legal communicator, even though the adversary can get intercepted transcripts.
- *Un-traceability.* The proposed scheme should provide untraceability to protect the irrelevance among users and drones. Any adversary cannot get users'(drones') behavior patterns and then trace them from the intercepted messages.
- *Session Key Agreement.* A session key will be established(shared) between users and drones for their further communication after executing the proposed scheme successfully. The others (e.g. legal user who does not participant this session, adversary) are unable to get any useful information from the session key.
- *Resistance against Various Attacks.* Generally, the proposed scheme should withstand impersonation attack, server spoofing attack, modification attack, drone capture attack, stolen smart device attack, replay attack, known session key attack and man-in-the-middle attack.

Table 1
Summary of notations.

Notation	Description
U_i, V_j	The ith user and jth drone, respectively
CS	Control server of the all users and drones
ID_i, ID_j	The identities of the ith user and jth drone
k, MSK	160 bits secret value and mask key of CS
n	160 bits public parameter selected by CS
$PID_{i,j,s}$	The pseudonym of U_i, V_j and CS, respectively
α_i, α_j	The master private key of U_i and V_j, respectively
r_1, r_2	160 bits random numbers of U_i and V_j, respectively
ST_1	The current timestamp
$\triangle T$	The maximum internal time threshold of accepting messages
$h(\cdot)$	Secure one-way hash function, where $h : \{0,1\}^* \rightarrow Z_n^*$
$\oplus$	Bitwise XOR operation
$\|$	Concatenation operation

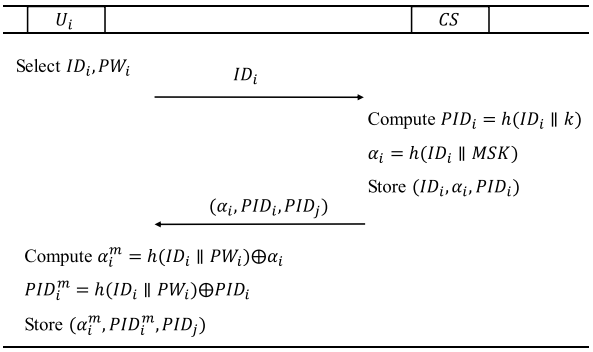


Fig. 3. User registration phase.

4. Proposed scheme

The proposed scheme is comprised of three parts: the setup phase, the registration phase and the mutual authentication phase. The notations used in this paper are defined as shown in Table 1.

4.1. Setup phase

In this phase, CS generates its master private key and other public system parameters in the following steps:

1. CS randomly chooses a 160 bits numbers MSK as its master private key, and then chooses a 160 bits mask key k and the public system parameter n.
2. CS chooses a secure one-way hash function $h : \{0,1\}^* \rightarrow Z_n^*$, its identity ID_s and computes $PID_s = h(ID_s \| k)$.
3. CS saves (MSK, k) secretly and publishes (h, n, PID_s).

4.2. User registration phase

In this phase, user U_i joins the IoD environment, registers on control server CS and gets his/her secret key via a secure channel. The computation steps are as shown in Fig. 3.

1. U_i first randomly selects his/her identity ID_i and password PW_i, then sends ID_i with registration request to CS.
2. Upon receiving the message from U_i, CS computes $PID_i = h(ID_i \| k)$, $\alpha_i = h(ID_i \| MSK)$ and stores (ID_i, α_i, PID_i) in list L_s securely. Then, CS sends (α_i, PID_i, PID_j) to U_i via a secure channel.
3. U_i receives (α_i, PID_i, PID_j) and computes $\alpha_i^m = h(ID_i \| PW_i) \oplus \alpha_i$, $PID_i^m = h(ID_i \| PW_i) \oplus PID_i$. Finally, U_i stores $(\alpha_i^m, PID_i^m, PID_j)$ securely.

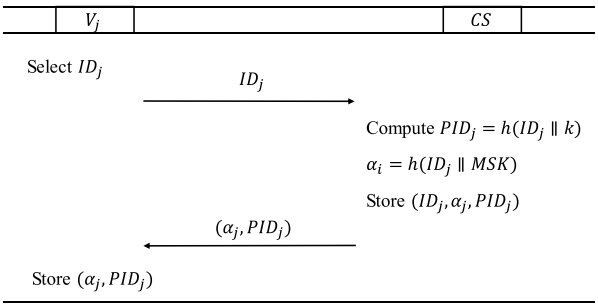


Fig. 4. Drone registration phase.

4.3. Drone registration phase

In this phase, Drone submits its identity to control server CS and get its secret key. The detailed steps are as shown in Fig. 4.

1. V_j randomly selects its identity ID_j and send it with registration request to CS.
2. CS computes $PID_j = h(ID_i \| k)$, $\alpha_j = h(ID_j \| MSK)$ and stores (ID_j, α_j, PID_j) in list L_s securely. Finally, CS sends (α_j, PID_j) to V_j via a secure channel.
3. V_j receives (α_j, PID_j) and stores them securely.

4.4. Authentication phase

U_i and V_j are two registered user and drone, respectively, after registration phase. They can communicate with each other securely after that U_i and V_j establish a session key. As in Fig. 5, U_i and V_j will do as follows.

1. U_i first inputs his/her identity ID_i and password PW_i, and the mobile will compute $PID_i = PID_i^m \oplus h(ID_i \| PW_i)$, $\alpha_i = \alpha_i^m \oplus h(ID_i \| PW_i)$. Then it randomly chooses a 160 bits number $r_1 \in Z_n^*$ and the current timestamp ST_1 to calculate the following. Finally, it sends authentication request message (M_1, M_2, M_3, M_4) to CS through a public channel.

$$M_1 = h(PID_s \| ST_1) \oplus PID_i$$
$$M_2 = h(PID_i \| PID_s \| \alpha_i) \oplus r_1$$
$$M_3 = h(PID_i \| PID_s \| \alpha_i \| r_1) \oplus PID_j$$
$$M_4 = h(PID_i \| PID_j \| PID_s \| \alpha_i \| r_1)$$

2. After receiving the authentication request message (M_1, M_2, M_3, M_4) from U_i, CS first checks the validation of time by $time - ST_1 \leq_\Delta T$, in which ΔT is the maximum time threshold of accepting messages and $time$ is the current time received message. If it is true, CS goes to the next step; Otherwise, CS rejects the authentication request. CS further computes $PID_i' = M_1 \oplus h(PID_s \| ST_1)$ and retrieves α_i' in the list L_s. Then CS computes the following.

$$r_1' = M_2 \oplus h(PID_i' \| PID_s \| \alpha_i')$$
$$PID_j' = M_3 \oplus h(PID_i' \| PID_s \| \alpha_i' \| r_1')$$
$$M_4' = h(PID_i' \| PID_j' \| PID_s \| \alpha_i' \| r_1')$$

3. CS checks the validation of $M_4' = M_4$. If they are equal, CS can authenticate U_i and retrieves α_j' in the list L_s through PID_j', then continue to do the following steps. Otherwise, CS rejects the authentication request. Finally, CS sends message (M_5, M_6, M_7) to V_j through a public channel.

$$M_5 = h(PID_j' \| \alpha_j') \oplus r_1'$$
$$M_6 = h(PID_j' \| PID_s \| \alpha_j' \| r_1') \oplus PID_i'$$
$$M_7 = h(PID_i' \| PID_j' \| PID_s \| \alpha_j' \| r_1')$$

4. After receiving message (M_5, M_6, M_7) from CS, V_j first computes the following:

$$r_1'' = M_5 \oplus h(PID_j \| \alpha_j)$$
$$PID_i'' = M_6 \oplus h(PID_j \| PID_s \| \alpha_j \| r_1'')$$
$$M_7' = h(PID_i'' \| PID_j \| PID_s \| \alpha_j \| r_1'')$$

5. V_j checks the validation of $M_7' = M_7$. If it does not hold, V_j rejects the communication request. Otherwise, V_j can authenticate CS and randomly choose a 160 bits number $r_2 \in Z_n^*$, then continue to do the following steps. Finally, V_j sends message (M_8, M_{10}) to U_i through a public channel.

$$M_8 = h(PID_j \| PID_i'' \| r_1'') \oplus r_2$$
$$M_9 = h(r_1'' \| r_2)$$
$$SK_{ji} = h(PID_i'' \| PID_j \| PID_s \| M_9)$$
$$M_{10} = h(PID_i'' \| PID_j \| PID_s \| r_1'' \| r_2 \| M_9)$$

6. When U_i receives message (M_8, M_{10}) from V_j, he/she first computes as the follows. U_i checks the validation of $M_{10}' = M_{10}$. If they are equal, U_i can authenticate V_j and calculate the common session key $SK_{ij} = h(PID_i \| PID_j \| PID_s \| M_9') = SK_{ji}$. Otherwise, U_i rejects the communication request.

$$r_2' = M_8 \oplus h(PID_j \| PID_i \| r_1)$$
$$M_9' = h(r_1 \| r_2')$$
$$M_{10}' = h(PID_i \| PID_j \| PID_s \| r_1 \| r_2')$$
$$SK_{ij} = h(PID_i \| PID_j \| PID_s \| M_9')$$

5. Security analysis

In this section, we analyze the security of the proposed scheme. First, we exhibit the proposed scheme is secure with the random oracle model. We then explain how the proposed scheme can satisfy the security requirements described in Section 5.3. In Section 5.4, we do a comparison of the proposed scheme and other two latest AKA schemes.

5.1. Security model

Based on Choi et al. [41], we propose a security model which is defined by a game played between an adversary $\mathcal{A}$ and a challenger C. The adversary $\mathcal{A}$ is simulated as Turing Machine which runs in probability polynomial time. The challenger C can simulate all the oracles. Π_Λ^t means the tth instance of the participants $\Lambda \in (U_i, CS, V_j)$. The oracle machines allow $\mathcal{A}$ to issue a series of queries adaptively to them and give the corresponding response.

- $h(x)$: The hash oracle maintains a hash list L_h. When $\mathcal{A}$ executes hash query with message x, C first check whether x is in the hash list L_h. If yes, C returns the result $h(x)$ to $\mathcal{A}$. Otherwise, C randomly chooses a number $X \in Z_n^*$, returns r to $\mathcal{A}$ and stores (x, X) in hash list L_h.
- $Extract(ID_i)$: The ability that $\mathcal{A}$ can corrupt a legal drone and obtain its secret key will be shown in this query. When $\mathcal{A}$ executes extract query on drone's identity ID_i, C returns the corresponding secret key to $\mathcal{A}$.
- $Send(\Pi_\Lambda^t, M)$: The ability that $\mathcal{A}$ can launch an active attack will be shown in this query. When $\mathcal{A}$ sends message M to instance Π_Λ^t, it will get the corresponding response from Π_Λ^t with message M. For a new instance Π_Λ^t, $\mathcal{A}$ can begin with sending $Send(\Pi_\Lambda^t, Start)$ to the oracle.
- $Reveal(\Pi_\Lambda^t)$: This query simulate the incorrect use of session key. When $\mathcal{A}$ executes this query, if the instance has been successfully produced, C will return the session key of instance Π_Λ^t. Otherwise, returns $\perp$.

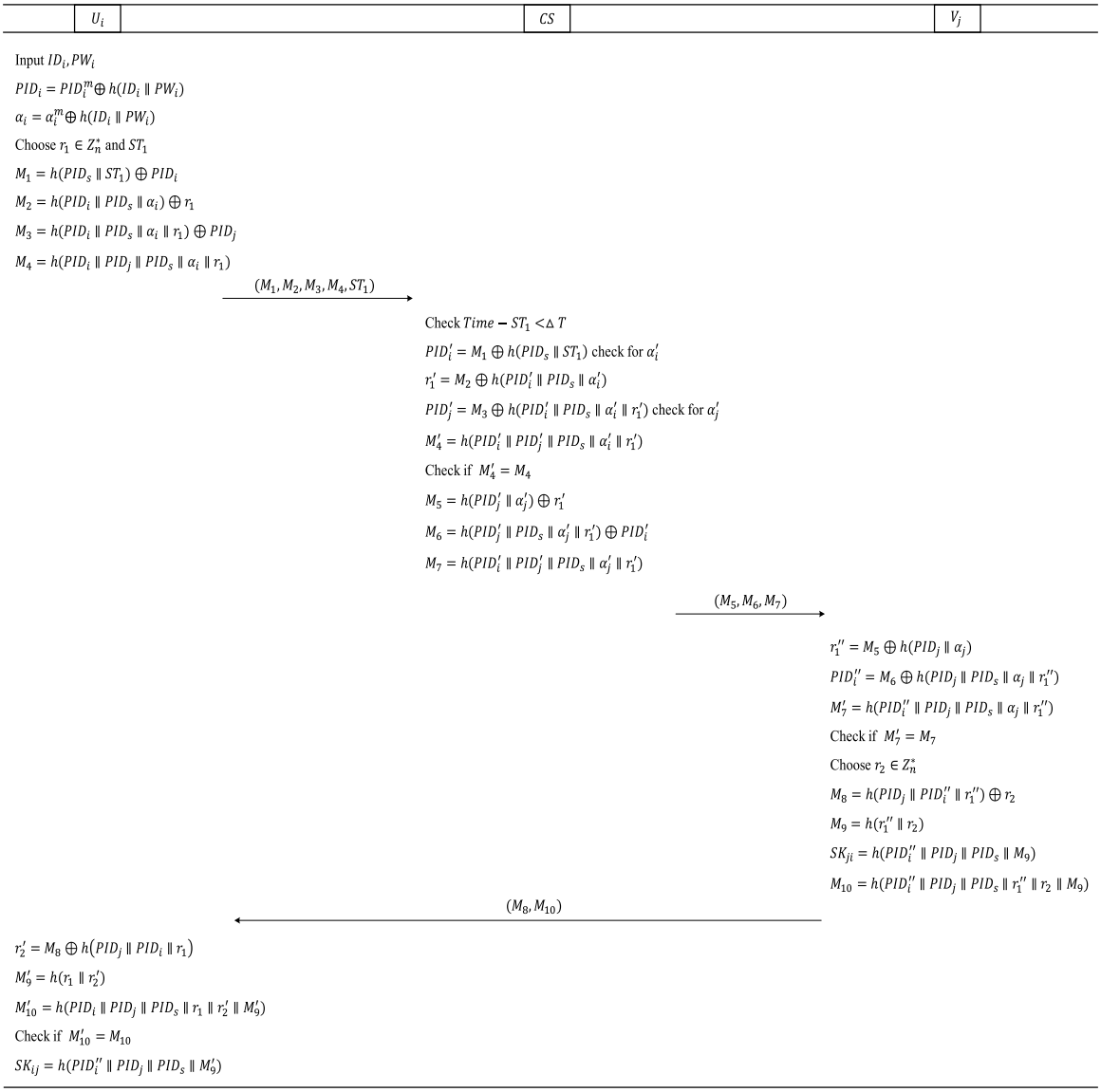


Fig. 5. Authentication and key agreement phase.

- $Execute(U_i, V_j)$: This query $\mathcal{A}$ can eavesdrop any messages in the public channel. When $\mathcal{A}$ executes this query, it can get all messages during the process.
- $Test(\Pi_A^t)$: This query $\mathcal{A}$ can distinguish between real session key and random secret key. $\mathcal{A}$ can execute this query only once. C randomly chooses a bit $b \in 0, 1$ and returns real session key to $\mathcal{A}$ if $b = 1$, otherwise, returns random secret key of the same size($b = 0$). In other case, if the queried instance Π_A^t does not have the session key, C returns $\perp$ to $\mathcal{A}$.

$\mathcal{A}$ can also continue to do $Extract, Send, Reveal, Execute$ queries after the $Test$ query. At this point, the limitation of $\mathcal{A}$ is that it cannot do $Reveal$ query for the oracle and its pattern oracle that has been executed $Test$ query.

At last, $\mathcal{A}$ outputs b' as a guess of b. We say $\mathcal{A}$ can successfully win this game (break the authentication and key agreement of the proposed scheme Σ) if $b' = b$. The advantage of $\mathcal{A}$ is defined as $adv_{\Sigma}^{AKA}(\mathcal{A}) = |2Pr[b' = b] - 1|$.

Define 1 (*AKA-Secure*)**.** If there is no probability polynomial adversary $\mathcal{A}$ can successfully win the game with non-negligible advantage $adv_{\Sigma}^{AKA}(\mathcal{A})$, we say the proposed scheme Σ is AKA-Secure.

$\mathcal{A}$ can successfully break the mutual authentication of the proposed scheme Σ, if $\mathcal{A}$ can forge a legal login message, a communication message or a response message. Let E_{U-CS} express the event that $\mathcal{A}$ impersonates the user U_i and generates a login message accepted by CS successfully. Let E_{U-V} express the event that $\mathcal{A}$ impersonates the drone V_j and generates a response message accepted by U_i. The advantage of $\mathcal{A}$ winning this game is defined as $adv_{\Sigma}^{MA}(\mathcal{A}) = Pr[E_{U-CS}] + Pr[E_{U-V}]$.

Define 2 (*MA-Secure*)**.** If there is no probability polynomial adversary $\mathcal{A}$ can successfully win the game with non-negligible advantage $adv_{\Sigma}^{MA}(\mathcal{A})$, we say the proposed scheme Σ is MA-Secure.

5.2. Provable security block

We prove that there is no adversary $\mathcal{A}$ can forge a legal login and response message in non-negligible probability. That means the proposed scheme is AKA-secure and MA-secure in the security block.

Lemma. *Assumption that the probability polynomial adversary $\mathcal{A}$ can calculate a legal login message or a response message with non-negligible probability. Then, there is a challenger C can guess 160 bits random number successfully with a non-negligible probability.*

Proof. C selects a 160 bits random number msk, and sends the parameters $\{h, n\}$ to $\mathcal{A}$. C generates a hash list L_h which is initially empty to record the inputs and outputs of the hash oracles, and selects two challenge drones' identities ID_I and ID_J at the beginning.

We suppose all the other oracles can be queried after the hash oracles are done. The answers to the queries are as follows:

- $h(x_i)$: C first checks whether x_i exists in the list L_h. If it exists, then C returns X_i to $\mathcal{A}$; if not, C randomly chooses a number X_i, adds (x_i, X_i) in list L_h and returns X_i to $\mathcal{A}$.
- $Extract(ID_i)$: If $i \neq I, J$, C seeks a tuple $(ID_i \parallel msk, \alpha_i)$ in the list L_h, and returns α_i to $\mathcal{A}$. Otherwise, C rejects the query and aborts the game.
- $Send(\Pi^t_A, M)$: $\mathcal{A}$ can lunch this query to simulate the active attack in four types.
  - $Send(\Pi^t_{U_i}, Start)$: C first checks whether $i \neq I$. C then seeks hash list L_s for U_i's secret key α_i if they are equal. With the help of secret key α_i, C chooses a random number $r_1 \in Z^*_n$, the current time-stamp ST_1 and computes (M_1, M_2, M_3, M_4). If they are not equal, C randomly selects three numbers $R_1, R_2, R_3 \in Z^*_n$ and sets $M_2 \leftarrow R_1$, $M_3 \leftarrow R_2$, $M_4 \leftarrow R_3$. Compute $M_1 = h(PID_s \parallel ST_1) \oplus PID_I$ and return (M_1, M_2, M_3, M_4) to $\mathcal{A}$.
  - $Send(\Pi^k_{V_j}, (M_5, M_6, M_7))$: On receiving the message, C first check whether j and J are equal. If yes, C casts away this message, selects two random numbers $R_4, R_5 \in Z^*_n$ and sets $M_8 \leftarrow R_4, M_{10} \leftarrow R_5$. Otherwise, C seeks hash list L_h for secret key α_j of V_j, and processes the scheme as usual.
  - $Send(\Pi^t_{U_i}, (M_8, M_{10}))$: C first checks whether$j \neq J$. C then seeks hash list L_s for V_j's secret key α_j if they are equal. With the help of secret key α_j, C chooses a random number $r_2 \in Z^*_n$, and computes (M_8, M_{10}). If they are not equal, C randomly selects three numbers $R_4, R_5, R_6 \in Z^*_n$, sets $r_2 \leftarrow R_4$ $M_8 \leftarrow R_5$, $M_{10} \leftarrow R_6$ and returns (M_8, M_{10}) to U_i.
- $Reveal(\Pi^t_A)$: If instance Π^t_A has been accepted, C returns its correct session key SK_A, otherwise, C returns $\perp$.

Assumption that the adversary $\mathcal{A}$ can calculate a legal login message or a response message successfully, that is to say, the answers (M_1, M_2, M_3, M_4) to $Send(\Pi^t_{U_i}, Start)$ query with $i = I$ and (M_8, M_{10}) to $Send(\Pi^k_{V_j}, (M_5, M_6, M_7))$ query with $j = J$ are passed the verification by CS and U_i. The following events are defined to calculate the advantage of C for convenience.

- E_1: The simulation is not aborted.
- E_2: $\mathcal{A}$ submits a legal login message (M_1, M_2, M_3, M_4) from $Send(\Pi^t_{U_i}, Start)$ query or a legal response message (M_8, M_{10}) from $Send(\Pi^k_{V_j}, (M_5, M_6, M_7))$ query, meanwhile, $Extract(ID_I)$ and $Extract(ID_J)$ have never been queried.
- E_3: $U_i = U_I$ or $V_j = V_J$.
- E_4: C can choose the correct tuples from hash list L_h.

Let q_s, q_{L_s} and q_{L_h} denote the number of $Send$-query, L_s-query and L_h-query executed by $\mathcal{A}$.

$$Pr[E_1] \geq \frac{1}{q_s}$$

It is obvious that

$$Pr[E_2|E_1] \geq \epsilon$$

$$Pr[E_3|(E_2 \wedge E_1)] \geq \frac{1}{q_{L_s}}$$

$$Pr[E_4|(E_3 \wedge E_2 \wedge E_1)] \geq \frac{1}{q_{L_s}} \frac{1}{q_{L_s} - 1} + \frac{a}{q_{L_h}} \frac{b}{q_{L_h} - a}$$

in which, a is the correct tuple number in $Send(\Pi^t_{U_i}, Start)$-query and b is the correct number of $Send(\Pi^t_{U_i}, (M_8, M_{10}))$-query. Therefore, the challenger C guesses the 160 bits random number successfully with the non-negligible probability as follows:

$$\begin{aligned} &Pr[E_1 \wedge E_2 \wedge E_3 \wedge E_4] \\ &= Pr[E_4|E_3 \wedge E_2 \wedge E_1] Pr[E_3|E_2 \wedge E_1] Pr[E_2|E_1] Pr[E_1] \\ &= \frac{1}{q_s} \frac{1}{q_{L_s}} \left(\frac{1}{q_{L_s}} \frac{1}{q_{L_s} - 1} + \frac{a}{q_{L_h}} \frac{b}{q_{L_h} - a}\right) \epsilon \end{aligned}$$

However, this is contradictions of the hardness of guessing the 160 bits random number. That means, $\mathcal{A}$ cannot generate a legal login message or a legal response message, and drones in the scheme can authenticate each other. □

Theorem 1. *The proposed scheme is MA-Secure under guessing 160 bits random number is hard.*

Form the Lemma, there is no $\mathcal{A}$ can generate a legal login message or a legal response message if it is hard to guess the 160 bits random number. Hence we can get the proposed scheme is MA-Secure.

Theorem 2. *The proposed scheme is AKA-Secure under guessing 160 bits random number is hard.*

Proof. Assumption that the probability polynomial adversary $\mathcal{A}$ outputs a correct $b' = b$ with non-negligible probability ϵ after executing $Test$-query. Then, there is a challenger C can guess 160 bits random number successfully with a non-negligible probability. The following events are defined to calculate the advantage of C for convenience.

- E_{SK}: $\mathcal{A}$ can get the correct session key after querying $Test$-query.
- E_U: $\mathcal{A}$ executes a $Test$-query to instance Π_{V_I} successfully.
- E_V: $\mathcal{A}$ executes a $Test$-query to instance Π_{V_J} successfully.
- E_{U-CS-V}: $\mathcal{A}$ can destroy the authentication between user and control server CS, and the authentication between U_i and V_j.

As we know the probability that $\mathcal{A}$ guesses a correct b without any other help information is 1/2, thus, we can get $Pr[E_{SK}] \geq \epsilon/2$. The following equation holds:

$$\begin{aligned} Pr[E_{SK}] &= Pr[E_{SK} \wedge E_V] + Pr[E_{SK} \wedge E_V \wedge E_{U-CS-V}] \\ &\quad + Pr[E_{SK} \wedge E_V \wedge \neg E_{U-CS-V}] \\ &\leq Pr[E_{SK} \wedge E_U] + Pr[E_{U-CS-V}] \\ &\quad + Pr[E_{SK} \wedge E_V \wedge \neg E_{U-CS-V}] \end{aligned}$$

Then we have

$$\begin{aligned} &Pr[E_{SK} \wedge E_U] + Pr[E_{SK} \wedge E_V \wedge \neg E_{U-CS-V}] \\ &\geq Pr[E_{SK}] - Pr[E_{U-CS-V}] \\ &\geq \epsilon/2 - Pr[E_{U-CS-V}] \end{aligned}$$

Owing to $Pr[E_V \wedge \neg E_{U-CS-V}] = Pr[E_V]$, thus

$$Pr[E_{SK} \wedge E_V] \geq \frac{\epsilon}{4} - \frac{Pr[E_{U-CS-V}]}{2}$$

The event $E_{SK} \wedge E_{V_i}$ shows $\mathcal{A}$ impersonates user U_i and gets the correct session key successfully. According to the Lemma, $Pr[E_{U-CS-V}]$ is a negligible probability, so that $\frac{\epsilon}{4} - \frac{Pr[E_{U-CS-V}]}{2}$ is non-negligible. That means the probability $\mathcal{A}$ can get the correct session key is non-negligible, this is contradictions of the hardness of guessing the 160 bits random number. □

5.3. Parameter analysis for security

In this subsection, we also exhibit that the proposed scheme satisfies the other security requirements described in Section 3.2.

Mutual Authentication: We can know that the advantage that $\mathcal{A}$ can forge the legal login message and response authentication message

is negligible, on the basis of Lemma in Section 5.2. Thus, U_i and V_j can authenticate each other with the aid of CS by verifying the validation of the transmitted messages. Therefore, the proposed scheme can achieve mutual authentication.

Anonymity: The user's identity ID_i is transmitted not directly in plain text but in a masked form, $PID_i = h(ID_i \| k)$ in our proposed scheme. Moreover, PID_i is embedded in $M_1 = h(PID_s \| ST_1) \oplus PID_i$. On account of the hardness of guessing 160 bits random number, the adversary $\mathcal{A}$ is infeasible to compute drone's real identity without knowing mask key k. Therefore, the proposed scheme can guarantee anonymity.

Un-traceability: In authentication phase, random nonces r_1, r_2, and current time-stamp are chosen in various sessions, so that the messages $(M_1, \ldots, M_{10})$ sent by the participant in every session are different. The adversary $\mathcal{A}$ cannot find the relationship among the messages sent by $U_i(CS/V_j)$ and also cannot trace the sender. Moreover, the real identities or pseudonyms $(ID_w, PID_w)_{w \in i,j,s}$ are not directly involved in messages but embedded in secure one-way collision-resistant hash function. Therefore, the proposed scheme can achieve un-traceability.

Session key Agreement: U_i authenticates V_j by checking the validation of M_{10} and V_j authenticates U_i by checking the validation of M_7, thus, U_i and V_j make sure they have the right random nonce r_1 and r_2. So, they can compute the session key $SK = SK_{ij} = SK_{ji} = h(PID_i \| PID_j \| PID_s \| h(r_1 \| r_2))$ and use the session key in the future communication. Therefore, the proposed scheme can provide secure session key agreement.

Resistance against Various Attacks: We will exhibit that our proposed scheme can withstand impersonation attack, server spoofing attack, modification attack, drone capture attack, stolen smart device attack, replay attack, known session key attack and man-in-the-middle attack. The detailed description are exhibited as follows.

- Impersonation Attack: Assume that the adversary $\mathcal{A}$ has captured a legal registered drone, so he knows all the secret information stored in drone. That is to say, $\mathcal{A}$ knows the pseudonyms of drones. Under the circumstances, $\mathcal{A}$ can impersonate U_i and V_j.
  - If $\mathcal{A}$ wants to impersonate a legal user U_i, he/she should generate the valid messages (M_1, M_4) and send them to CS. Given that $\mathcal{A}$ knows user's pseudonym accidentally. $\mathcal{A}$ computes a valid $M_1 = h(PID_s \| ST_1) \oplus PID_i$ and $M_4 = h(PID_i \| PID_j \| PID_s \| \alpha_i^* \| r_1)$, where r_1 and α_i^* are randomly selected by $\mathcal{A}$ as U_i's random nonce and secret key. Upon receiving the message (M_1, M_4), CS first parses PID_i from M_1 and retrieve the corresponding secret key α_i in list L_s. Then CS computes M_4' with α_i and checks whether M_4' is equal to M_4. However, $\mathcal{A}$ does not know the real α_i, thus, CS can distinguish the impersonated U_i from real user.
  - If $\mathcal{A}$ wants to impersonate a legal drone V_j, he/she should generate the valid messages M_{10} and send it to U_i. $\mathcal{A}$ randomly select r_1^* and r_2 and computes $M_{10} = h(PID_i \| PID_j \| PID_s \| r_i^* \| r_2 \| h(r_i^* \| r_2))$. Upon receiving the message M_{10}, U_i calculates M_{10}' with real random number r_1 and checks whether M_{10}' is equal to M_{10}. However, $\mathcal{A}$ does not know the real r_1, thus, U_i can distinguish the impersonated V_j from real drone.
- Server Spoofing Attack: $\mathcal{A}$ pretends itself as the control server and sends a legal message M_7 to V_j. $\mathcal{A}$ computes $M_7 = h(PID_i \| PID_j \| PID_s \| \alpha_j^* \| r_1)$, where α_j^* is a random number selected as V_j's secret key by $\mathcal{A}$. On receiving the message M_7, V_j calculates M_4' with α_j and checks whether M_7' is equal to M_7. However, $\mathcal{A}$ cannot get real α_j, thus, V_j can find out the vicious server. Therefore, the proposed scheme can resist the server spoofing attack.
- Modification Attack: We can see that the transmitted messages $M_4(M_7)$ is composed of sender's(receiver's) secret key $\alpha_i(\alpha_j)$, on the basis of Theorems 1 and 2. $CS(V_j)$ can estimate whether the message is modified by checking the equation of $M_4 = M_4'(M_7 = M_7')$. Besides, M_{10} contains receiver's random number r_1, U_i can detect any modification of M_{10} by checking the equation of $M_{10} = M_{10}'$. Therefore, the proposed scheme can resist modification attack.
- Drone Capture Attack: As we have presented in Sections 1 and 2, drones are vulnerable. Suppose $\mathcal{A}$ has captured c drones and gets their stored and communication information: $\alpha_j = h(ID_j \| MSK)$, $PID_j = h(ID_j \| k)$, $SK_{ij} = h(PID_i \| PID_j \| PID_s \| h(r_1 \| r_2))$, $j \in (1, \ldots, c)$. The master key MSK and mask key k are embedded in secure one-way hash function, thus, even though $\mathcal{A}$ gets $3c$ information, it cannot calculate the correct master key MSK and mask key k. Since the session key is compromised of pseudonyms and random numbers, $\mathcal{A}$ cannot compute the next communication session key without knowing the random numbers. Therefore, the proposed scheme can resist drone capture attack.
- Stolen Smart Device Attack: If $\mathcal{A}$ steals user's smart device and extracts the stored data $(\alpha_i^m, PID_i^m, PID_j)$ through side channel attack, in which $\alpha_i^m = \alpha_i \oplus h(ID_i \| PW_i)$ and $PID_i^m = PID_i \oplus h(ID_i \| PW_i)$. The adversary can guess user's password PW_i, however, he cannot verify the correctness without knowing user's identity ID_i. Therefore, the proposed scheme can resist stolen smart device attack.
- Replay Attack: Both two entities choose random numbers $(r_1, r_2 \in Z_n^*)$ and calculates the login message M_4 and the response message M_{10}. Owing to the freshness of r_1 and r_2, CS, V_j and U_i can distinguish the replayed message from the received messages by checking the validation of them. Therefore, the proposed scheme can resist the replay attack.
- Known Session Key Attack: If the adversary $\mathcal{A}$ knows the session key for a particular session. As we know, the session key SK is a hash value of participants' pseudonyms and the random numbers. On account of the collision-resistant secure one-way hash function, $\mathcal{A}$ cannot parse the random numbers from SK. And for the other sessions, $\mathcal{A}$ cannot compute the right session key without knowing current random numbers. Therefore, the proposed scheme can resist known session key attack.
- Man-In-The-Middle Attack: From Section 5.2, we can see that U_i can be authenticated by CS through its secret key, V_j can authenticate CS because CS knows its secret key and V_j can be identified by U_i with the help of the knowing of r_1. Thus, all the participants can authenticate each other. Therefore, the proposed scheme can resist man-in-the-middle attack.

5.4. Security comparisons

The comparison of security requirements between the proposed scheme and two latest lightweight authentication schemes [15,16] designed for devices in Internet of Things is provided in Table 2. In Wazid et al.'s scheme [15], if adversary is a legal user, he can know all pseudonyms of registered users and get what he needs to calculate SK from message Msg_3. Thus, this scheme cannot provide session key agreement. In Singh et al.'s scheme [16], the adversary can calculate nodes' secret values S_i, S_j from the transmitted messages, and then impersonate the nodes. Our proposed scheme can satisfy all the security requirements, and have better security than the other two schemes.

6. Performance evaluation

In this section, we demonstrate the performance of our proposed scheme in terms of communication costs and computation costs, and we also compare the results with Wazid et al.'s scheme [15] and

Table 2
Comparison of security requirements.

Requirements	Ref. [15]	Ref. [16]	Our scheme
Mutual authentication	Yes	No	**Yes**
Anonymity	Yes	No	**Yes**
Un-traceability	Yes	No	**Yes**
Session key agreement	No	No	**Yes**
Impersonation attack	Yes	No	**Yes**
Server spoofing attack	Yes	–	**Yes**
Modification attack	Yes	No	**Yes**
Drone capture attack	Yes	No	**Yes**
Stolen smart device attack	Yes	–	**Yes**
Replay attack	Yes	Yes	**Yes**
Known session key attack	Yes	Yes	**Yes**
Man-in-the-middle attack	Yes	No	**Yes**

Table 3
Executing time of various operations (MS).

Operations	User (Drone) side	Server side
T_f	13.405	5.427
T_h	0.056	0.007
T_{exp}	2.249	0.339
T_{mul}	0.008	0.001

Singh et al.'s scheme [16]. We show the executing time of performing various operations in our proposed scheme, Wazid et al.'s scheme and Singh et al.'s scheme. The following symbols are used to represent the executing time in this paper.

- T_f: Time to perform a fuzzy extraction.
- T_h: Time to perform a secure hash function.
- T_{exp}: Time to perform a modular exponentiation.
- T_{mul}: Time to perform a modular multiplication.

On the basis of executing time used in He et al.'s scheme [42], the above operations are implemented between a mobile(drone) device and a desktop computer. The drones are equipped with the same as mobile devices (e.g. camera, microphone, infrared, biochemical detector), therefore, we consider the drone as a mobile device. The mobile (drone) device is simulated on Samsung Galaxy S5, which has a Quad-core 2.45G processor, 2G bytes memory and the Android 4.4.2 operation system. The server is simulated on a desktop computer, which has I5-4460S 2.90 GHz processor, 4G bytes memory and the window 8 operating system.

To achieve the security level of 1024 bits RSA algorithm, we choose a multiplicative cyclic group G with the order of n, which is a 160 bits prime number. The executing time of this operations is listed in Table 3.

In Wazid et al.'s scheme, the user side needs to execute sixteen hash functions and one fuzzy extraction. Therefore, the user's executing time is $1T_f + 16T_h$ about 14.301 ms. The drone side calls for executing seven hash functions and the server side requires to calculate eight hash functions. That means, the executing time of drone side and server side are $7T_h$ about 0.392 ms and $8T_h$ about 0.056 ms respectively. In Singh et al.'s scheme, the server side is not involved in the authentication phase. Thus, in authentication phase, the user side needs to execute two exponentiation functions and five modular multiplication operations. Therefore, the executing of user side is $2T_{exp}$+$5T_{mul}$ about 4.538 ms. The drone side requires to calculate two exponentiation functions and seven modular multiplication operations. Therefore, the drones' executing time is $2T_{exp} + 7T_{mul}$ about 4.554 ms.

In our proposed scheme, the operation used in this authentication phase is only hash function. From the authentication phase, we can get the user side, drone and server should calculate ten hash functions, seven hash functions and seven hash functions, respectively. That is to say, the executing time of user, drone and server are $10T_h$, $7T_h$ and $7T_h$, about 0.56, 0.392 and 0.049 ms. We compare the computation cost of our scheme with Wazid et al.'s scheme and Singh et al.'s scheme. The computation cost of the three schemes is shown in Table 4 and Fig. 6. We note the hash function is more suitable for drone environment.

Table 4
Comparison of computation cost.

Scheme	User side	Drone side	Server side	Total
Ref. [15]	$1T_f + 16T_h$(14.301)	$7T_h$(0.392)	$8T_h$(0.056)	14.794
Ref. [16]	$2T_{exp} + 5T_{mul}$(4.538)	$2T_{exp} + 7T_{mul}$(4.554)	–	9.092
Our scheme	$10T_h$(0.56)	$7T_h$(0.392)	$7T_h$(0.049)	1.001

Table 5
Comparison of communication cost.

Scheme	No. of messages	Communication cost	Length (bits)
Ref. [15]	3	$10\|Z_n\| + 3\|ID\|$	1696
Ref. [16]	2	$4\|G\| + 4\|ID\|$	4256
Our scheme	3	$9\|Z_n\| + \|ID\|$	1472

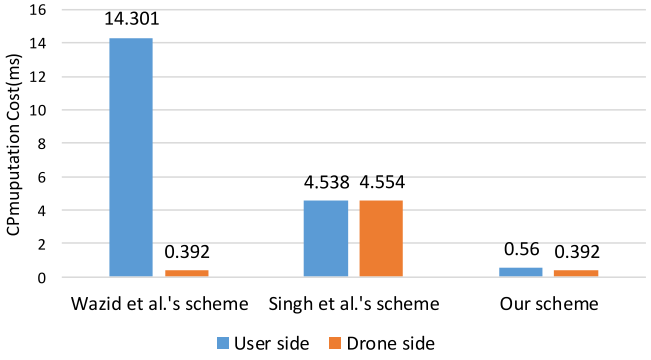


Fig. 6. Comparison of computation cost.

Let $|G|$ represents the 1024 bits length of element in G and $|Z_n|$ denotes the 160 bits length of the element in Z_n. The symbol $|ID|$ means 32 bits length of time-stamp and user' identity. We compare the communication cost of our scheme with Wazid et al.'s scheme and Singh et al.'s scheme. The transmitted messages in Wazid et al.'s scheme are $(M_1, \ldots, M_7, M_{10}, M_{11}, M_{12}, T_1, T_2, T_3)$, in which T_i is a 32 bits time-stamp, $M_i \in Z_n$. Thus, the total communication cost is $10|G| + 3|ID|$ about 1696 bits. The communication transcripts in Singh et al.'s scheme are $(X_i, Y_i, Time_i, ID_i$ from user side and $X_j, Y_j, Time_j, ID_j)$ from drone side, in which $Time_i$ is a 32 bits time-stamp and ID_i is a 32 bits user's identity. The total communication cost is $4|G| + 4|ID|$ about 4256 bits.

In our proposed scheme, the user should send $(M_1, \ldots, M_4, ST_1)$ to server, the server will send (M_5, M_6, M_7) to drone, and the drone calculates and sends (M_8, M_{10}) back to user. ST_1 is the 32 bits time-stamp, all the other messages are 160 bits hash values. Thus, the total communication cost is $9|Z_n| + |ID|$ about 1472 bits. The comparison of these three schemes is shown in Table 5 and Fig. 7. Our proposed scheme has lower communication overhead than Wazid et al.'s scheme and Singh et al.'s scheme.

7. Conclusion

The applications in IoD architecture have been widely used in various fields and brought a great convenience from military to civilian. In the last years, several authentication schemes for IoD have been proposed. However, most of them are subjected to serious security risks and have high communication and computation cost. We design a lightweight AKA scheme between drones and users with the help of the server. Our proposed scheme can be proven secure under random oracle model, and it also can achieve the security requirements of the IoD environment and withstand various attacks. In addition, the comparisons of communication and computation cost show that our proposed scheme has better performance.



Fig. 7. Comparison of communication cost.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The data used to support the findings of this study are available from the corresponding author upon request.

Acknowledgments

This work is partially supported by the National Key Research and Development Program of China (No. 2018YFC1315404), the National Natural Science Foundation of China (Nos. 61972294, 61932016), the Opening Project of Guangdong Provincial Key Laboratory of Data Security and Privacy Protection, China (No. 2017B030301004-11) and the Science and Technology planning project of ShenZhen, China (No. JCYJ20170818112550194).

References

[1] M. Gharibi, R. Boutaba, S.L. Waslander, Internet of Drones, IEEE Access 4 (2016) 1148–1162, http://dx.doi.org/10.1109/ACCESS.2016.2537208.

[2] Y.-J. Chen, L.-C. Wang, Privacy protection for Internet of Drones: A network coding approach, IEEE Internet Things J. 6 (2) (2019) 1719–1730, http://dx.doi.org/10.1109/JIOT.2018.2875065.

[3] S. Aggarwal, N. Kumar, Path planning techniques for unmanned aerial vehicles: A review, solutions, and challenges, Comput. Commun. 149 (2020) 270–299, http://dx.doi.org/10.1016/j.comcom.2019.10.014, URL http://www.sciencedirect.com/science/article/pii/S0140366419308539.

[4] R. Valentino, W.-S. Jung, Y.-B. Ko, A design and simulation of the opportunistic computation offloading with learning-based prediction for unmanned aerial vehicle (UAV) clustering networks, Sensors 18 (11) (2018) 3751.

[5] M. Erdelj, B. Uk, D. Konam, E. Natalizio, From the eye of the storm: An IoT ecosystem made of sensors, smartphones and UAVs, Sensors 18 (11) (2018) 3814.

[6] B. Vergouw, H. Nagel, G. Bondt, B. Custers, Drone technology: Types, payloads, applications, frequency spectrum issues and future developments, in: The Future of Drone Use, Springer, 2016, pp. 21–45, http://dx.doi.org/10.1007/978-94-6265-132-6_2.

[7] S. Saharan, S. Bawa, N. Kumar, Dynamic pricing techniques for intelligent transportation system in smart cities: A systematic review, Comput. Commun. 150 (2020) 603–625, http://dx.doi.org/10.1016/j.comcom.2019.12.003, URL http://www.sciencedirect.com/science/article/pii/S0140366419310990.

[8] N. Kumar, N. Chilamkurti, J.J.P.C. Rodrigues, Learning automata-based opportunistic data aggregation and forwarding scheme for alert generation in vehicular ad hoc networks, 39 (3) (2014) 22–32.

[9] M. Bae, H. Kim, Authentication and delegation for operating a multi-drone system, Sensors 19 (9) (2019) 2066.

[10] R. Kaur, N. Kumar, S. Batra, Trust management in social Internet of Things: A taxonomy, open issues, and challenges, Comput. Commun. 150 (2019) http://dx.doi.org/10.1016/j.comcom.2019.10.034.

[11] C.-T. Li, C.-C. Lee, C.-Y. Weng, A secure chaotic maps and smart cards based password authentication and key agreement scheme with user anonymity for telecare medicine information systems, J. Med. Syst. 38 (9) (2014) 77.

[12] C.-T. Li, C.-C. Lee, C.-Y. Weng, C.-I. Fan, An extended multi-server-based user authentication and key agreement scheme with user anonymity, KSII Trans. Internet Inf. Syst. 7 (1) (2013).

[13] Y.-J. Chen, L.-C. Wang, Privacy protection for Internet of Drones: A network coding approach, IEEE Internet Things J. 6 (2) (2018) 1719–1730.

[14] A.S. Sohal, R. Sandhu, S.K. Sood, V. Chang, A cybersecurity framework to identify malicious edge device in fog computing and cloud-of-things environments, Comput. Secur. (2018) S0167404817301827.

[15] M. Wazid, A.K. Das, N. Kumar, A.V. Vasilakos, J.J.P.C. Rodrigues, Design and analysis of secure lightweight remote user authentication and key agreement scheme in Internet of Drones deployment, IEEE Internet Things J. 6 (2) (2019) 3572–3584, http://dx.doi.org/10.1109/JIOT.2018.2888821.

[16] J. Singh, A. Gimekar, S. Venkatesan, An efficient lightweight authentication scheme for human-centered industrial Internet of Things, Int. J. Commun. Syst. (2019) e4189, http://dx.doi.org/10.1002/dac.4189.

[17] L. Lamport, Password authentication with insecure communication, Commun. ACM 24 (11) (1981) 770–772.

[18] J.H. Cheon, K. Han, S.-M. Hong, H.J. Kim, J. Kim, S. Kim, H. Seo, H. Shim, Y. Song, Toward a secure drone system: Flying with real-time homomorphic authenticated encryption, IEEE access 6 (2018) 24325–24339.

[19] P. Gope, T. Hwang, An efficient mutual authentication and key agreement scheme preserving strong anonymity of the mobile user in global mobility networks, J. Netw. Comput. Appl. 62 (2016) 1–8.

[20] P. Gope, T. Hwang, Lightweight and energy-efficient mutual authentication and key agreement scheme with user anonymity for secure communication in global mobility networks, IEEE Syst. J. 10 (4) (2015) 1370–1379.

[21] C. Wang, Y. Zhu, W. Shi, V. Chang, P. Vijayakumar, B. Liu, Y. Mao, J. Wang, Y. Fan, A dependable time series analytic framework for cyber-physical systems of IoT-based smart grid, ACM Trans. Cyber-Phys. Syst. 3 (2018) 1–18, http://dx.doi.org/10.1145/3145623.

[22] M. Turkanović, B. Brumen, M. Hölbl, A novel user authentication and key agreement scheme for heterogeneous ad hoc wireless sensor networks, based on the Internet of Things notion, Ad Hoc Netw. 20 (2014) 96–112.

[23] M.S. Farash, M. Turkanović, S. Kumari, M. Hölbl, An efficient user authentication and key agreement scheme for heterogeneous wireless sensor network tailored for the Internet of Things environment, Ad Hoc Netw. 36 (2016) 152–176.

[24] R. Amin, S.H. Islam, G. Biswas, M.K. Khan, L. Leng, N. Kumar, Design of an anonymity-preserving three-factor authenticated key exchange protocol for wireless sensor networks, Comput. Netw. 101 (2016) 42–62.

[25] Q. Jiang, S. Zeadally, J. Ma, D. He, Lightweight three-factor authentication and key agreement protocol for internet-integrated wireless sensor networks, IEEE Access 5 (2017) 3376–3392.

[26] S. Challa, M. Wazid, A.K. Das, N. Kumar, A.G. Reddy, E.-J. Yoon, K.-Y. Yoo, Secure signature-based authenticated key establishment scheme for future IoT applications, IEEE Access 5 (2017) 3028–3043.

[27] S.D. Selvi, S. Vivek, C. Rangan, Certificateless KEM and hybrid signcryption schemes revisited, in: International Conference on Information Security Practice and Experience, Springer, 2010, pp. 294–307.

[28] F. Li, M. Shirase, T. Takagi, Certificateless hybrid signcryption, in: International Conference on Information Security Practice and Experience, Springer, 2009, pp. 112–123.

[29] D. He, J. Chen, J. Hu, A pairing-free certificateless authenticated key agreement protocol, Int. J. Commun. Syst. 25 (2) (2012) 221–230.

[30] M. Geng, F. Zhang, Provably secure certificateless two-party authenticated key agreement protocol without pairing, in: 2009 International Conference on Computational Intelligence and Security, Vol. 2, IEEE, 2009, pp. 208–212.

[31] G. Yang, C.-H. Tan, Strongly secure certificateless key exchange without pairing, in: Proceedings of the 6th ACM Symposium on Information, Computer and Communications Security, ACM, 2011, pp. 71–79.

[32] H. Sun, Q. Wen, H. Zhang, Z. Jin, A novel pairing-free certificateless authenticated key agreement protocol with provable security, Front. Comput. Sci. 7 (4) (2013) 544–557.

[33] S.-H. Seo, J. Won, E. Bertino, pCLSC-TKEM: a pairing-free certificateless signcryption-tag key encapsulation mechanism for a privacy-preserving IoT, Trans. Data Priv. 9 (2) (2016) 101–130.

[34] J. Won, S.-H. Seo, E. Bertino, Certificateless cryptographic protocols for efficient drone-based smart city applications, IEEE Access 5 (2017) 3721–3749.

[35] C.T. Li, C.C. Lee, C.Y. Weng, A secure chaotic maps and smart cards based password authentication and key agreement scheme with user anonymity for telecare medicine information systems, J. Med. Syst. 38 (9) (2014) 77.

[36] C.C. Lee, Y.M. Lai, C.T. Chen, S.D. Chen, Advanced secure anonymous authentication scheme for roaming service in global mobility networks, Wirel. Pers. Commun. 94 (3) (2016) 1–16.

[37] P. Vijayakumar, V. Chang, L.J. Deborah, B. Balusamy, P.G. Shynuc, Computationally efficient privacy preserving anonymous mutual and batch authentication schemes for vehicular ad hoc networks, Future Gener. Comput. Syst. 78 (2016) 943–955.

[38] R. Amin, S.H. Islam, P. Vijayakumar, M.K. Khan, V. Chang, A robust and efficient bilinear pairing based mutual authentication and session key verification over insecure communication, Multimedia Tools Appl. 77 (13) (2017) 1–26.

[39] D. He, Y. Zhang, D. Wang, K.-K.R. Choo, Secure and efficient two-party signing protocol for the identity-based signature scheme in the IEEE P1363 standard for public key cryptography, IEEE Trans. Dependable Secure Comput. 1 (99) (2018) 1–10, http://dx.doi.org/10.1109/TDSC.2018.2857775.

[40] Q. Feng, D. He, S. Zeadally, N. Kumar, K. Liang, Ideal lattice-based anonymous authentication protocol for mobile devices, IEEE Syst. J. 13 (3) (2018) 2775–2785, http://dx.doi.org/10.1109/JSYST.2018.2851295.

[41] K.Y. Choi, J.Y. Hwang, D.H. Lee, I.S. Seo, ID-based authenticated key agreement for low-power mobile devices, in: Australasian Conference on Information Security and Privacy, Springer, 2005, pp. 494–505.

[42] D. He, S. Zeadally, N. Kumar, W. Wu, Efficient and anonymous mobile user authentication protocol using self-certified public key cryptography for multi-server architectures, IEEE Trans. Inf. Forensics Secur. 11 (9) (2016) 2052–2064.



Yunru Zhang received her Bachelor and Master degree in cyber science and engineering from Wuhan University, Wuhan, China, in 2015 and 2018, respectively. She is currently purse her Ph.D degree in the School of Cyber Science and Engineering, Wuhan University, China. Her main research interests include cryptography and information security, in particular, authentication key agreement, cryptographic protocols and zero-knowledge proof.



Debiao He received his Ph.D. degree in applied mathematics from School of Mathematics and Statistics, Wuhan University in 2009. He is currently an Professor of the School of Cyber Science and Engineering, Wuhan University. His main research interests include cryptography and information security, in particular, cryptographic protocols.



Li Li received her Ph.D degree in computer science from Computer School, Wuhan University. She is currently an associate professor at School of Software, Wuhan University. Her research interests include data security and privacy, applied cryptography and security protocols.



Biwen Chen received his M.S. degree from Hubei University of Technology of China in 2016. He is currently pursing his Ph.D. degree at Computer School, Wuhan University, Wuhan, China. His main research interests include cryptography and information security, in particular, cryptographic protocols.